

PROG1

Computer Programming 1

Grade 12, Semester 1 study outline

A summary of section headings and key terms, made by students for revision. It is not the handout and does not replace it.

01. Introduction to Computers and Programming

I. HUMBLE BEGINNINGS

"The simplicity of ancient devices is as amazing as the complexity of modern day machines. Times have changed

1. Responds to a specific set of instructions in a well-defined manner
2. Executes a prerecorded list of instructions

The devices that we have now are mostly, if not all, electronic or powered by electricity but computing

II. GENERATIONS OF COMPUTERS

These generations describe the major technological developments that truly changed the way computers

III. REAL-WORLD PROGRAMMING APPLICATIONS

The applications of computer programming are everywhere. Any piece of hardware, software, and

02. Fundamental Programming Concepts

I. REVIEW: COMPUTER

The modern-day definition of a computer is that it is an electronic machine that is programmed to both

II. KEY PROGRAMMING TERMINOLOGIES

In its raw form, it is a set of instructions written in code (keywords and symbols) that a computer

1. Low-level Languages - These include the Machine Language and Assembly Language

Machine language or machine code is the default language of a computer.

III. ALGORITHM

It is a set of instructions (but not yet in pure code form) for solving a specific problem. It is a "well-

1. Prepare an empty glass (Glass C)
2. Pour Glass A content (Coke) into Glass C
3. Pour Glass B content (Sprite) to Glass A
4. Pour the Glass C content (Coke) into Glass B
5. Glass A now contains Sprite, and Glass B contains Coke

glassC = glassA //Glass C gets the content of Glass A

Key terms

He - someone who runs a program. Unlike the programmer, he/she doesn't have to understand

Although Ada Lovelace - first person to have written a programmable algorithm, the word "algorithm"

Natural language - not the best way for writing program algorithms, as it is too wordy and can be

03. Program Design Process

I. ALGORITHMIC MINDSET

"Not everyone needs to be a coder. The idea of coding is valuable and wonderful, but the idea everyone should

1. Prepare a temporary variable
2. Get input value for 1st variable
3. Get input value for 2nd variable
4. Temporary variable takes the value of the 1st variable
5. 1st variable takes the value of the 2nd variable

6. 2nd variable takes the value of the temporary variable

7. Display the new values of the variables

II. PHASES OF PROGRAM DESIGN

In designing a program, it is customary for a programmer to undergo two major phases to ensure that the

1. Programmer's Documentation

It contains the technicalities including the algorithms used, source code with comments, test

2. User's Documentation

This is a guide for the end-user. It consists of instructions for installing the program as well as a

Key terms

Whether the program you - creating is simple or complex, the characteristics mentioned above should be

04. Basic Elements of Programming

1. Data

It refers to any value that the computer gets as an input, stores for future use, and transforms into a

2. Variable

It is a specific area/location in the computer memory where a value/data is stored and retrieved. There

3. Input

This is the process of getting data from the programmer or user to be processed by the computer. Basic

4. Output

This is the process of producing processed data. An example is printing out something on the terminal

5. Processes/Operations

What lie between or connect the input and output elements are the processes that are running inside a

1. The variable "num1" has an initial value of 5

2. "num1" "takes the value of" "num1" (which is 5) plus 5

3. Therefore, the new value of num1 is 10

Additionally, when a variable passes through an operation, it is possible that its value will not

1. The variable "num1" has an initial value of 5

2. The variable "num2" has an initial value of 10

3. "num2" "takes the value of" "num1" (which is 5) plus 15

6. Subroutines/Functions

There are functions (portions of codes that can be executed independently) that are needed by the

Key terms

Pi - always 3.14159... and other values that a programmer will define to remain the same all throughout the

05. Logical-Control Structures in Programming

1. Sequence Logic / Sequential Flow

This is the default control structure. As the name suggests, it is the execution of code line by line. The

2. Selection Logic / Conditional Flow

This is the structure that enables the program to select or choose between two or more flows. This is also

3. Iteration Logic / Repetitive Flow

It is a programming structure that allows line(s) of code to be repeated or executed multiple times as

Key terms

Control structures - considered as the "building blocks" of computer programs. They specify the flow of
Different statements - performed, depending on which side of the condition is met (true or false)

06. Types of Operators Part 1

1. Assignment Operator

This operator denoted by '=' (equal) assigns/sets a value to an empty variable or changes/resets the

1. num1 is assigned the value of 5

2. num2 is assigned the value of 10

3. num3 is assigned the value of the sum of num1 and num2

In conjunction with the Arithmetic Operators (which will be discussed next), here are other ways of using

2. Arithmetic Operators

These are operators that carry out computations.

3. Increment and Decrement Operators

Increment and decrement, denoted by "++" and "--" respectively, are special types of unary

Key terms

Multiplication and Division - at the same "priority level" as addition and Subtraction. In other words, if the

07. Types of Operators Part 2

4. Relational Operators

As the name implies, these operators test or define the kind of relation between two or more entities. They

5. Logical Operators

They are used to combine relational operators to make more complex decisions. Common logical operators

6. Other Operators

They are still other operators which programming languages such as C++ support such as Bitwise Operators

08. Flowcharting Part 1-Symbols and Usage

1. Terminal / Terminator Block

It marks the start or end of a program or process. It usually contains the word START/BEGIN or STOP/END.

2. Flow Line

It is used to show the flow or direction of tasks in a process or program. It shows the relationship between

3. Preparation / Initialization / Declaration Block

It is used to declare (tell the computer of) and/or initialize variables needed for the program. In actual

4. Input / Output Block

It is used for getting inputs and displaying outputs. Its usage depends on the description inside the shape. For

5. Process Block

It is used to indicate a process step (instructions or actions). Here you will apply the assignment, arithmetic,

6. Decision / Condition Block

It is used to represent decision-making statements. Here you will apply the relational and logical

7. Predefined Process Block

It is used to call a specific function or subprogram (independent portions of a code or flowchart) - review

8. On-page Connector

It is used to connect blocks on the same page. It marks a jump between different sections of the flow in one

9. Off-page Connector

It is used to connect blocks that are located on a different page. Off-page connectors are usually named

10. Annotation

It represents comments about certain portions of the flowchart. They have no effect on the behavior of the

09. Flowcharting Part 2-General Rules and Tracing

I. GUIDELINES FOR FLOWCHARTING

A flowchart is an effective way of showing steps in a process or program but it can also get confusing and

1. Follow Proper Direction

Structure your flowchart from top to bottom or left to right (with the exception of looping/iterative parts)

2. Use Correct Symbols

You have learned from the previous lesson that each symbol has a specific function. While it may seem

3. Use Flow Lines (Arrows) Not Lines

All the blocks in a flowchart should be connected using flow lines. Using just lines (no arrow heads)

4. Apply Proper Naming Conventions for Variables

Though you are not doing actual coding yet, it is good to already practice proper use of variable names. In

5. Ensure Entry and Exit Points on Symbols

By default, flowchart symbols can have an entry point on the top or left side of the shape (except for a

6. Keep Everything on One Page

If possible, keep the flowchart of a particular program fit on a single page while making sure that the texts

7. Use Line Jumps when Necessary

If you have flow lines that cross each other, use line jumps (a.k.a. edge jumps or bridges). Many tools used in

8. Trace Your Flowchart

To ensure that every flow leads to the right direction, every condition properly filters values, and expected

II. TRACING

Flowchart Tracing means simulating the execution of a program on paper (you may also use a document or

Key terms

Though you - not doing actual coding yet, it is good to already practice proper use of variable names. In

Flowchart Tracing - simulating the execution of a program on paper (you may also use a document or

10. Initialization, End-User's View, and Input Validation

No outline could be read from this handout.

11. Basic Calculator Algorithm

No outline could be read from this handout.

12. The Real Number System

I. REAL NUMBER

Mathematics and Computer Programming are intertwined. If you want to master the latter, you should also

II. POSITIVE and NEGATIVE NUMBERS

Positive numbers are integers that are greater than zero (right side of zero in the number line) while

III. EVEN and ODD NUMBERS

Even numbers are integers that are divisible by 2. They end with digits 0, 2, 4, 6, or 8 while odd numbers are

Key terms

Mathematics and Computer Programming - intertwined. If you want to master the latter, you should also

Positive numbers - integers that are greater than zero (right side of zero in the number line) while

Even numbers - integers that are divisible by 2. They end with digits 0, 2, 4, 6, or 8 while odd numbers are

13. Multiple and Nested Conditions

I. MULTIPLE and NESTED CONDITIONS

In real life, there are situations where we need to make some decisions based on certain scenarios or

II. CONDITION-DEPENDENT OUTPUTS

Conditional statements help route the program to produce outputs that are based on certain criteria. For

Key terms

Video games - great example of the use of conditional statements. In programming, a conditional

Minecraft - another game which is a perfect example of a program that uses a multiple condition structure

14. Number Sequences

I. NUMBER SEQUENCE DEFINED

A number series/sequence is a progression or list of numbers that are linked by a certain logical rule or

II. COMMON TYPES OF NUMBER SEQUENCES

1. Arithmetic Sequence

It is formed by adding the same value each time. When creating an arithmetic number sequence you

2. Geometric Sequence

It is formed by multiplying the same value each time. When creating a geometric number sequence you

3. Incremental Sequence

This is formed by adding an "increasing" number as you continue the number sequence.

4. Triangular Number Sequence

This is formed by following a pattern of dots that form a triangle. By adding another row of dots and

5. Square Sequence

This is formed by calculating the squares of consecutive counting numbers starting from 1.

6. Cube Sequence

This is formed by calculating the cubes of consecutive counting numbers starting from 1.

7. Fibonacci Sequence

It is a unique sequence that was said to have been discovered or "invented" by Leonardo Fibonacci, an

III. FLOWCHARTING NUMBER SEQUENCES

Given the descriptions about number sequences, let us create flowcharts that can easily generate some of

15. Design Patterns in Programming

I. "STAR" PATTERNS/DESIGNS

If in the previous lesson, you learned about the importance of number sequences in sharpening one's

II. FLOWCHARTING DESIGNS

You have learned in Lesson 5 the three major structures in programming namely, the Sequential,

Key terms

CLI - one the common interfaces (how a user interacts with the computer) used to see the outputs of

16. Measures of Central Tendency

I. MEASURES OF CENTRAL TENDENCY DEFINED

The "measures of central tendency" (also known as measures of center or central location) is a key concept

1. Mean

It is the sum of all values divided by the total number of values. It is also known as the arithmetic

2. Median

It is the middle value in an ordered/sorted list of numbers. It is the measure that divides the population into

3. Mode

It is the most common number in the set of data or the value that occurs most often. It is possible to have

II. FLOWCHARTING THE ARITHMETIC MEAN

Programming the mean, median, and mode would be best done using the concept of arrays (to be

17. Introduction to Arrays

I. DEFINITION OF ARRAYS

An array is a collection of elements of the same data type that are stored in adjacent memory locations,

1. integral (char, short, int, long, and bool)

2. enum

3. floating (float, double, long double)

1. array

2. struct

3. union

4. class

There are two types of arrays. A one dimensional array has its components arranged in list form. A

II. REAL-WORLD APPLICATIONS OF ARRAYS

Think of a cabinet. It is basically one object, stored in one specific place in your house (maybe in your room),

III. USING ARRAYS IN PROGRAMMING

Most, if not all, computer games that are programmed using C++ are utilizing arrays. The concept of arrays is

IV. FLOWCHARTING ARRAYS

Example #1: Create a flowchart that will get 50 names of students and store them in an array.

Key terms

Initializing - putting initial values to an array. For example, num[10] is an array with 10

18. Overview of C++ Programming

I. BACKGROUND

C++ is a general-purpose programming language that was developed as an enhancement of or successor to

II. C++ BASICS

In order to code and execute a C++ source code, you need to have a basic knowledge of the C++ syntax, just

1. Single Line Comment //this is a comment

2. Multi-line/Block Comment /*this is a

Usually, there are introductory comments in a source code to give a brief description of what the whole

III. SOME SIMPLE C++ PROGRAMS

Example 1: Create a program that will ask for the user's name, age, and email address then display them.